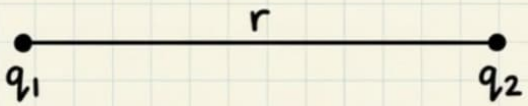
Coulomb's Law

This law was stated by French physicist Charles Augustin Coulomb.
It states that the force between 2 point charges, say q_1 & q_2, is given by:



$$F \propto q_1 \cdot q_2$$

$$F \propto \frac{1}{r^2}$$

Combining, we get:

$$F \propto \frac{q_1 \cdot q_2}{r^2}$$

or $$\boxed{F = k\frac{q_1 q_2}{r^2}}$$

where $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ C^2/m^2$

ϵ_0 → Permittivity of free space/vacuum $= 8.85 \times 10^{-12}\ C^2/N \cdot m$

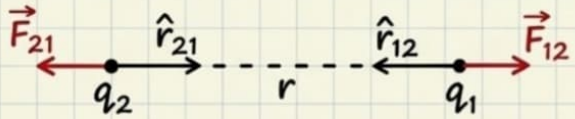
When there is some other medium between charges, Dielectric Constant (K) or Relative Permittivity (ϵ_r) is used:

$$K = \frac{\epsilon}{\epsilon_0}$$

$$F_m = \frac{1}{4\pi\epsilon_0 K} \cdot \frac{q_1 q_2}{r^2} \Rightarrow F_m = \frac{F_v}{K}$$

Vector Form of Coulomb's Law

Consider 2 charges q_1 & q_2 separated by a fixed distance r apart.
Here, $q_1, q_2 > 0$ (both are positively charged).



$$F_{12} = k \cdot \frac{q_1 q_2}{r^2} \hat{r}_{21} \quad \text{--- (1)}$$

$$F_{21} = k \cdot \frac{q_1 q_2}{r^2} \hat{r}_{12} \quad \text{--- (2)}$$

From eq 1 & 2, since $\hat{r}_{12} = -\hat{r}_{21}$

$$\Rightarrow F_{12} = -F_{21}$$

Conclusions:
1. Follows Newton's 3rd Law of Motion.
2. Electrostatic forces are central forces (act from center of 1 body to center of another).

Electric Field

Refers to the force per unit charge. Calculated by a test charge q_0 placed near q.

$$F = k \cdot \frac{q \cdot q_0}{r^2} \qquad \therefore E = \frac{F}{q_0}$$

$$E = \frac{k \cdot q}{r^2}$$

Electric field is always of the single e^-/obj.

Continuous Charge Densities

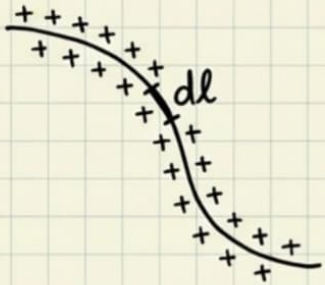
What if charges are not point-like, but instead distributed over a body?

Line Charge Density (λ)

Charge per unit length.

In $d\ell$ length, charge = dq.

$\therefore\ dq = \lambda \cdot d\ell$

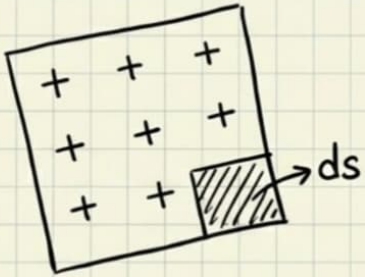


Surface Charge Density (σ)

Charge per unit area.

In ds area, charge = dq.

$\therefore\ dq = \sigma \cdot ds$



Volume Charge Density (ρ)

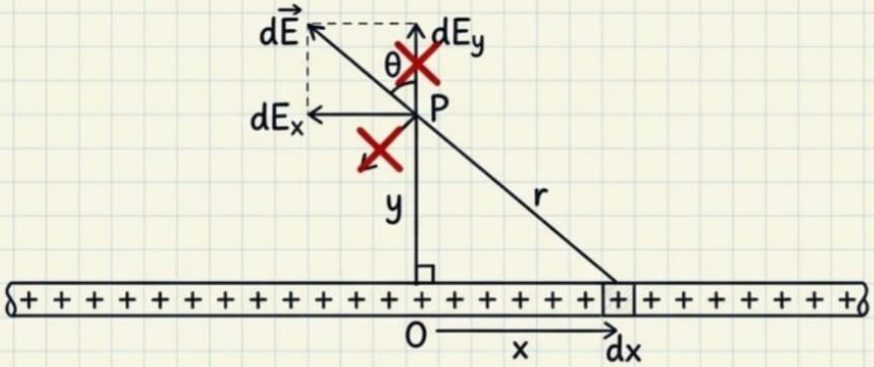
Charge per unit volume.

In dV volume, charge = dq.

$\therefore\ dq = \rho \cdot dV$

Electric Field of a Line Charge (Setup & Resolution)

Let us consider a uniform line charge of density λ. Point O is at the mid of the wire. Point P is at 'y' distance perpendicular to the wire. Current element dx is at 'x' distance from O. Distance from dx to P is 'r'.



$$dE = \frac{k \cdot dq}{r^2}$$

where $dq = \lambda \cdot dx$

In ΔPOQ, by Pythagoras theorem:

$$r^2 = x^2 + y^2$$

$$\therefore dE = \frac{k \cdot \lambda \cdot dx}{x^2 + y^2}$$

Vector Resolution

Field has 2 components:

$dE_x = -dE \cos\theta$ (negative as it points left direction)

$dE_y = dE \sin\theta$

Due to symmetry from an element on the opposite side, dE_y components will cancel out, but dE_x components will add up.

$\therefore dE_{net} = 2\, dE_x = 2\, dE \cos\theta$

Electric Field of a Line Charge (Integration)

Integrating both sides: $E = 2\int_0^{\infty} \frac{k\lambda\, dx \cdot \cos\theta}{x^2 + y^2}$

$$E = 2k\lambda \int_0^{\infty} \frac{\cos\theta \cdot dx}{x^2 + y^2}$$

$$\Downarrow$$

Substitution Trick: $\begin{cases} \text{Let } x = y\tan\theta \\ \text{Differentiating w.r.t } \theta \Rightarrow dx = y\sec^2\theta\, d\theta \end{cases}$

Limit Shift: When $x = 0$, $\theta = 0$
When $x = \infty$, $\theta = \pi/2$

$$\Downarrow$$

Substituting back in: $E = 2k\lambda \int_0^{\pi/2} \frac{\cos\theta \cdot y\sec^2\theta\, d\theta}{y^2(1 + \tan^2\theta)}$

$$\Downarrow$$

Simplifying, since $(1 + \tan^2\theta) = \sec^2\theta$: $E = \frac{2k\lambda}{y} \int_0^{\pi/2} \cos\theta\, d\theta$

$$\Downarrow$$

$$E = \frac{2k\lambda}{y} [\sin\theta]_0^{\pi/2}$$

$$E = \frac{2k\lambda}{y} [1 - 0]$$

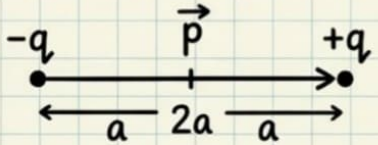
$$\Downarrow$$

$$\boxed{E = \frac{2k\lambda}{y}}$$

The Electric Dipole & Axial Point Field

Electric Dipole

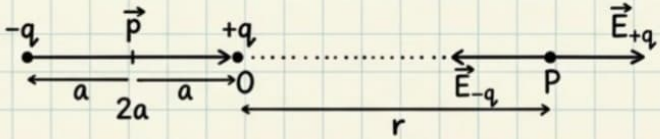
It is a set up of 2 equal but opposite charges, separated by a fixed distance.



Electric Dipole Moment (p): A moment developed from (-q) to (+q).

$p = q \cdot 2a$. SI unit is C.m (e.g., H_2O, HCl, NaCl).

At Axial Point



$$E_{+q} = \frac{kq}{(r-a)^2}\hat{p} \text{ (towards right)}$$

$$E_{-q} = \frac{kq}{(r+a)^2}(-\hat{p}) \text{ (towards left)}$$

$$E_{net} = E_{+q} + E_{-q}$$

$$E_{net} = kq\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right]\hat{p}$$

Simplifying...

$$E_{net} = kq\left[\frac{(r+a)^2 - (r-a)^2)}{(r-a)^2(r+a)^2}\right]\hat{p}$$

$$E_{net} = kq\left[\frac{4ar}{(r^2-a^2)^2}\right]\hat{p}$$

$$E_{net} = \frac{2 \cdot p \cdot r \cdot k}{(r^2-a^2)^2}\hat{p}$$

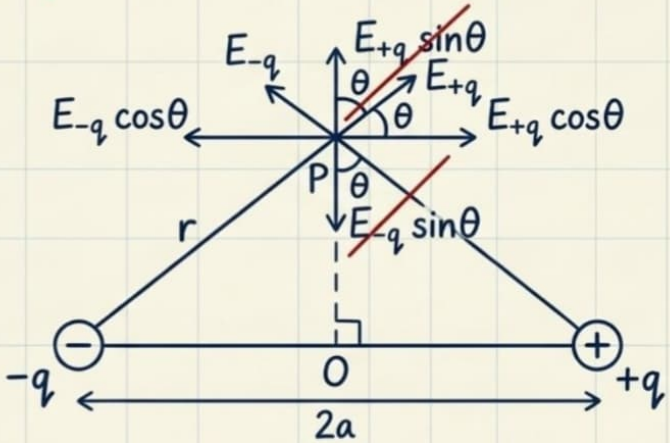
Special Case: If $r \gg a$, $a^2 \approx 0 \Rightarrow E_{net} = \frac{2kp}{r^3}\hat{p}$

Dipole Field: Equatorial Point

Let point P be perpendicular to midpoint O at distance r.

Differentiating electric fields into components:

The sin components will cancel out each other, and cos components will add up.



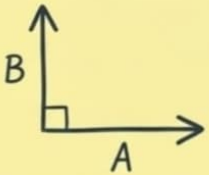
$$E_{net} = -2E_{+q}\cos\theta(\hat{p}) \; ; \quad E_{net} = -2\left(\frac{kq}{r^2+a^2}\right)\left(\frac{a}{(r^2+a^2)^{1/2}}\right)$$

Simplifying using $\vec{p} = 2qa$: $E_{net} = \dfrac{-\vec{p}\cdot\vec{k}}{(r^2+a^2)^{3/2}}\hat{p}$

Special Case: For $r \gg a \Rightarrow E_{equatorial} = \dfrac{-k\cdot\vec{p}}{r^3}\hat{p}$

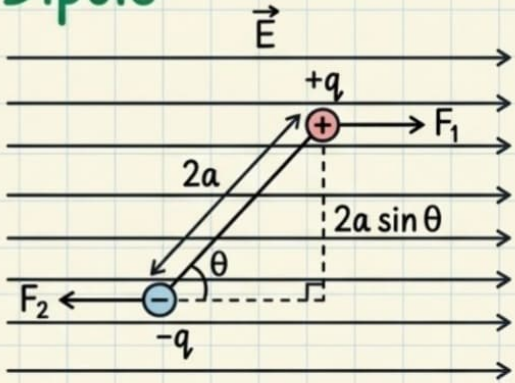
Side-by-Side Insight:
Axial magnitude is exactly 2× Equatorial magnitude for short dipoles!

Special Cases for finding resultants

Case 1 : $\theta = 0°$		$R = \sqrt{A^2 + B^2 + 2AB(1)}$ $R = \sqrt{(A+B)^2}$ $\therefore R = A + B$
Case 2 : $\theta = 180°$		$R = \sqrt{A^2 + B^2 + 2AB(-1)}$ $R = \sqrt{(A-B)^2}$ $\therefore R = A - B$
Case 3 : $\theta = 90°$		$R = \sqrt{A^2 + B^2 + 0} \Rightarrow R = \sqrt{A^2 + B^2}$ Direction : $\alpha = \tan^{-1}\left(\frac{B}{A}\right)$

Torque on an Electric Dipole

Let us consider a dipole of 2 charges $+q$ & $-q$ separated by a distance $2a$, placed in an electric field $\vec{E}$ at an angle θ.



Force on $+q \Rightarrow F_1 = +qE$ (along the field)
Force on $-q \Rightarrow F_2 = -qE$ (opposite to field)
Net force on dipole $= F_1 + F_2 = +qE - qE = 0$.

Now, Torque on the electric dipole is given as: $\tau = \text{Force} \times \text{Perpendicular distance}$

$$\tau = qE \times 2a \sin\theta$$

$$\tau = E \times 2aQ \times \sin\theta$$

Since dipole moment $p = 2aq$, substitute it in: $\tau = pE \sin\theta$

Vector Form: $\vec{\tau} = \vec{p} \times \vec{E}$

Properties of Electric Lines of Force

Origins & Terminations

1. Smooth, continuous curves without breaks.
2. Start at +ve charges & end at −ve charges. Cannot form closed loops.
9. Do not pass through a conductor ($\because \vec{E}$ inside charged conductor is 0).

Interactions & Geometry

3. Tangent to line gives direction of $\vec{E}$ at that point.
4. No 2 lines can cross. Reason: If they intersect, there will be 2 tangents at the point of intersection & hence 2 directions of $\vec{E}$ at the same pt, which is not possible.

5. Always normal to the surface of a conductor in equilibrium. Reason: If not normal, the parallel component would cause e^- to move and set up a surface current.

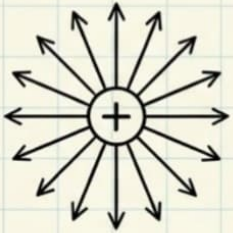
Behavioral Tendencies & Strength

6. Contract lengthwise → Explains attraction between 2 unlike charges.
7. Expand laterally → Explains repulsion between 2 similar charges.
8. Relative closeness gives measure of strength:
   (i) Close together in a strong field.
   (ii) Far apart in weak field.
   (iii) Parallel/equally spaced in a uniform field.

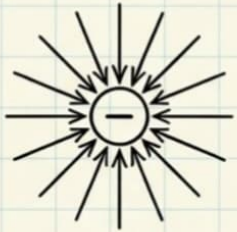
Field Topographies & Electric Flux

Visual Atlas of Field Lines

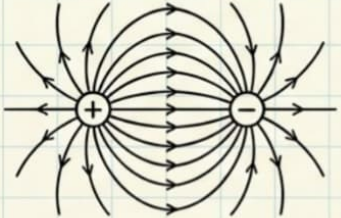
Positive Point Charge



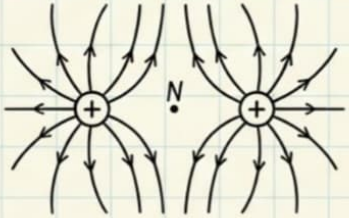
Negative Point Charge



Two Equal & Opposite Charges

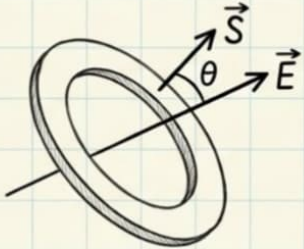


Two Equal & Same Charges



Electric Flux (ϕ)

It is the no. of Electric Field Lines passing perpendicularly through an area.



θ = angle between $\vec{E}$ & Area Vector $\vec{S}$

If $\vec{E}$ & $\vec{S}$ are parallel, $\theta = 0°$

$$\phi = \vec{E} \cdot \vec{S} \cos\theta$$

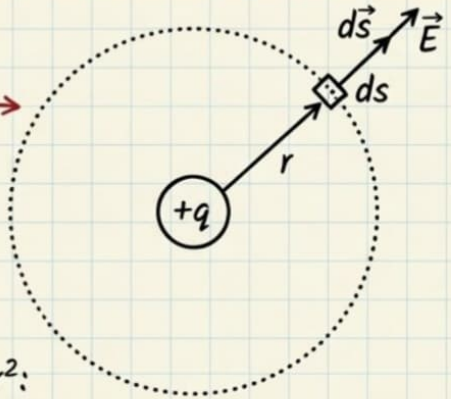
$$\phi = E \cdot S \cos 0° = E \cdot S$$

Gauss's Theorem (The Elegant Shortcut)

It states that total flux through a closed surface is $\frac{1}{\epsilon_0}$ × the charge enclosed in it.

Let us consider a +ve isolated charge q, surrounded by a Gaussian sphere of radius r.

Let ds be a small area element.



Notice the angle between ds & $\vec{E}$ is 0° everywhere on the sphere!

$$d\phi = E \cdot ds$$

$$d\phi = E \cdot ds \cdot \cos 0° = E \cdot ds$$

Integrating both sides,

$$\phi = E \cdot S$$

Substitute point charge field $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$ and sphere area $S = 4\pi r^2$:

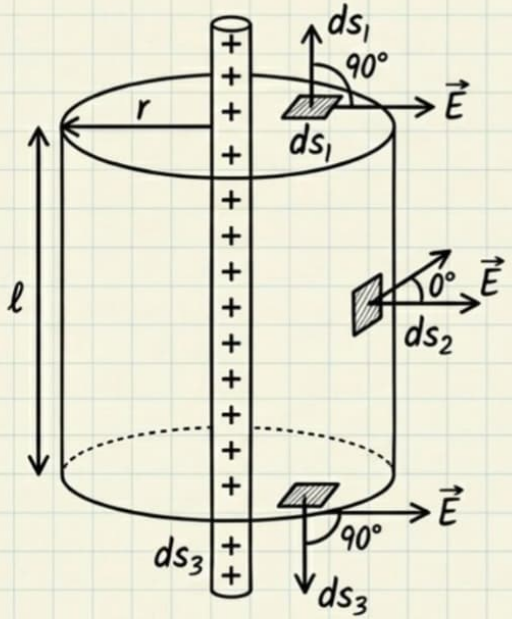
$$\phi = \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\cancel{r^2}}\right) \cdot \left(\cancel{4\pi r^2}\right)$$

Result:

$$\phi = \frac{q_{encl}}{\epsilon_0}$$

Application I: Infinite Line Charge (Gauss Shortcut)

Synthesis Note: Remember the 2-page integration trick we did earlier? Gauss solves this exact problem in 4 lines!



Let us consider a conductor carrying charge of length 'l' & a Gaussian surface around it of radius r.

Total electric flux is given by:

$$\phi = \int E\, ds_1 \cos 90° + \int E\, ds_2 \cos 0° + \int E\, ds_3 \cos 90°$$

$$\phi = 0 + \int E\, ds_2 + 0$$

$$\phi = E \int ds_2 = E \cdot \vec{S}_2$$

Since $S_2 = 2\pi r l$ here:

$$\phi = E \cdot 2\pi r l \quad \text{--- (1)}$$

Also, by Gauss Law: $\phi = \dfrac{q_{enclosed}}{\epsilon_0} = \dfrac{\lambda l}{\epsilon_0}$ --- (2)

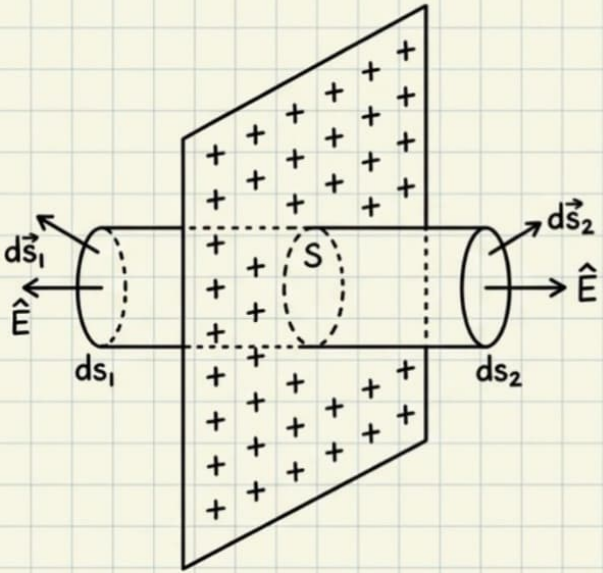
Equating (1) & (2):

$$E \cdot 2\pi r l = \frac{\lambda l}{\epsilon_0}$$

Result:

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$$

Application 2: Infinite Plane Sheet



Let us consider a plane sheet of area S with uniform charge density σ.

Draw a cylindrical Gaussian surface across the plane sheet.

Total electric flux is given by: $\phi = \oint \vec{E}\cdot d\vec{s}_1 + \oint \vec{E}\cdot d\vec{s}_2$

$$\phi = \oint E\, ds_1 \cos 0^\circ + \oint E\, ds_2 \cos 0^\circ$$

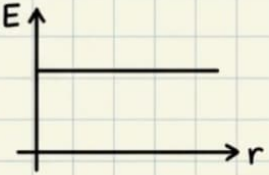
$$\phi = E\oint ds_1 + E\oint ds_2$$

Since $S_1 = S_2 = S$: $\phi = ES + ES = 2ES$

Also, $\phi = \dfrac{q_{encl}}{\epsilon_0} = \dfrac{\sigma S}{\epsilon_0}$ --- (1)

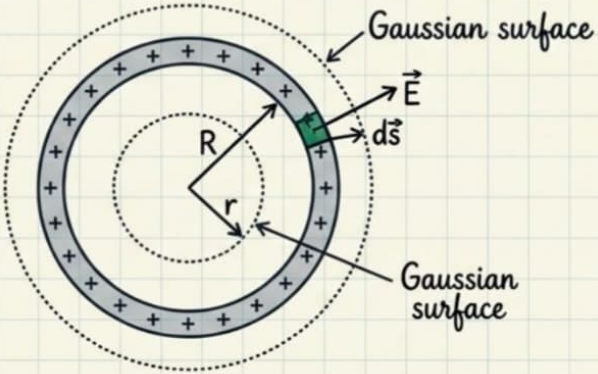
Equating flux equations: $\dfrac{\sigma S}{\epsilon_0} = 2ES$

$\hookrightarrow$ $E = \dfrac{\sigma}{2\epsilon_0}$



E is completely independent of r.

Application 3: Spherical Shell



Region 1: Outside ($r > R$)

Total electric flux: $\phi = \oint \vec{E} \cdot d\vec{s} = E \oint ds \cos 0^\circ$

$= E(4\pi r^2)$

Also, $\phi = \dfrac{q}{\epsilon_0}$

Equating: $E(4\pi r^2) = \dfrac{q}{\epsilon_0}$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} = \frac{kq}{r^2}$$

Region 2: On the Surface ($r = R$)

Substitute $r = R$ and $q_{encl} = \sigma \cdot 4\pi R^2$

$$\phi = E(4\pi R^2) = \frac{\sigma \cdot 4\pi R^2}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{\epsilon_0}$$

Region 3: Inside ($r < R$)

$q_{encl} = 0$ (charges reside only on the outer surface).

$$\phi = E(4\pi r^2) = 0$$

$$\Rightarrow E = 0$$